

OBGEL Model Sheet: Cu–Ag – Liquid

Open Binary Gibbs Energy Library

v0.2.0 candidate

Phase

The Liquid phase is represented as a substitutional solution model.

Composition

The independent composition variable is $x = x_{\text{Cu}}$ with $x_{\text{Ag}} = 1 - x$.

Molar Gibbs free energy

$$\begin{aligned} G_m(x, T) = & xG_{\text{Cu}}^0(T) + (1 - x)G_{\text{Ag}}^0(T) \\ & + RT [x \ln x + (1 - x) \ln(1 - x)] \\ & + x(1 - x) \sum_k L_k(T) (2x - 1)^k. \end{aligned}$$

Redlich–Kister parameters

$$\begin{aligned} L_0(T) &= 17384.4 - 4.46438T, \\ L_1(T) &= -1660.74 + 2.31516T. \end{aligned}$$

Standard-state Gibbs energies

The pure-component functions are piecewise functions from the SGTE data for pure elements. The complete OBGEL model is limited by the common documented range of these functions.

Cu in Liquid

$$\begin{aligned} G_{\text{Cu}}^0(T) = & 5194.277 + 120.97333T - 24.112392T \ln T - 0.00265684T^2 \\ & + 1.29223 \times 10^{-7}T^3 + 52478T^{-1} - 5.849 \times 10^{-21}T^7 & 298.15 \leq T \leq 1357.77 \\ & - 46.545 + 173.88148T - 31.38T \ln T & 1357.77 \leq T \leq 3200 \end{aligned}$$

Ag in Liquid

$$\begin{aligned} G_{Ag}^0(T) = & 3815.564 + 109.31099T - 23.846331T \ln T - 0.001790585T^2 \\ & - 3.98587 \times 10^{-7}T^3 - 12011T^{-1} \\ & - 1.034 \times 10^{-20}T^7 & 298.15 \leq T \leq 1234.93 \\ & - 3587.111 + 180.96466T - 33.472T \ln T & 1234.93 \leq T \leq 3000 \end{aligned}$$

Source and validation

The binary interaction parameters are reproduced from Kusoffsky (2002), which attributes them to a 1998 private communication from H. L. Lukas. NIST identifies the Hayes–Lukas–Effenberg–Petzow assessment as the full Cu–Ag thermodynamic assessment. Independent numerical evaluation of the complete Cu–Ag model gives a eutectic temperature of approximately 1053 K (779.9 C), consistent with the NIST value of 779.1 C.